Loan Default Analysis & Risk Assessment

Power BI Data Analytics Project Report

1. Executive Summary

The objective of this project is to analyze **loan default patterns and financial risk factors** using an interactive **Power BI dashboard**. The analysis focuses on identifying key borrower characteristics that contribute to loan default risk and providing insights that can help financial institutions improve **credit risk management and lending decisions**.

The dashboard integrates borrower **financial data, demographic attributes, and risk indicators** to provide a comprehensive view of loan performance.

Through this analysis, several important patterns were identified related to **income levels, demographic characteristics, and financial risk scores**, enabling organizations to make **data-driven lending decisions and reduce potential credit losses**.

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2. Business Problem

Financial institutions face significant losses due to **loan defaults**. Identifying high-risk borrowers before loan approval is critical.

The key business questions addressed in this project include:

- What factors contribute to **loan defaults**?
- Which **borrower profiles** are most likely to default?
- How do **income, financial status, and demographics** affect loan repayment?
- Can borrowers be segmented into **risk categories** for better credit decision making?

The goal is to create a **data-driven framework** that helps lenders reduce default rates while improving loan approval strategies.

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3. Dataset Overview

The dataset used in this project contains borrower information including:

Borrower Demographics

- Age
- Employment status
- Education level
- Marital status
- Residential status

Financial Attributes

- Annual Income

- Loan Amount
- Credit Score
- Debt-to-Income Ratio
- Existing Liabilities

Loan Information

- Loan Status
- Loan Purpose
- Default Indicator

These variables allow analysts to explore **relationships between borrower characteristics and loan repayment behavior**.

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4. Data Preparation & Modeling

Before building the dashboard, several **data preparation steps** were performed:

Data Cleaning

- Removed missing or inconsistent records
- Standardized financial values
- Verified categorical data consistency

Data Modeling

The data was structured into a **relational model** to support efficient analysis.

Typical relationships include:

Borrower Table
Loan Details Table
Financial Indicators Table

This structure allows the Power BI model to support **multi-dimensional analysis** across borrower characteristics.

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5. Key Analytical Measures (DAX KPIs)

In loan risk analytics, the dashboard typically uses several **calculated measures** to evaluate financial performance.

Total Loan Applications

Represents the total number of borrowers who applied for loans.

Total Loan Amount

Total capital distributed by the lending institution.

Default Loan Count

Total number of loans classified as default.

Default Rate

Percentage of loans that resulted in default.

Default Rate is one of the **most important KPIs** for risk evaluation.

Average Borrower Income

Helps identify the financial capability of borrowers.

Average Loan Size

Measures the average value of loans issued.

Risk Category Distribution

Segments borrowers into **low risk, medium risk, and high risk groups**.

These measures provide the foundation for the **visual analytics presented in the dashboard**.

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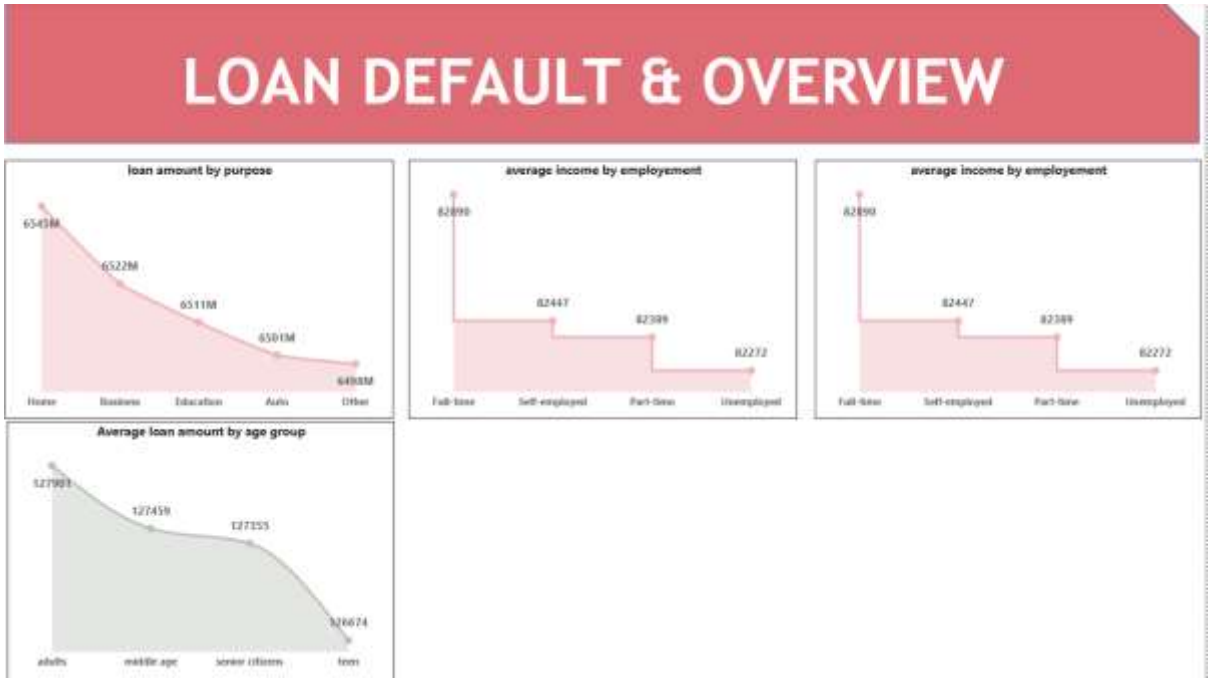
6. Dashboard Page Analysis

Page 1: Default Loan Overview



This page provides a **high-level overview of loan performance** and default statistics.

Key Insights

- Displays the **overall loan distribution**
- Shows the **proportion of default vs non-default loans**

- Highlights total loan amount issued by the institution

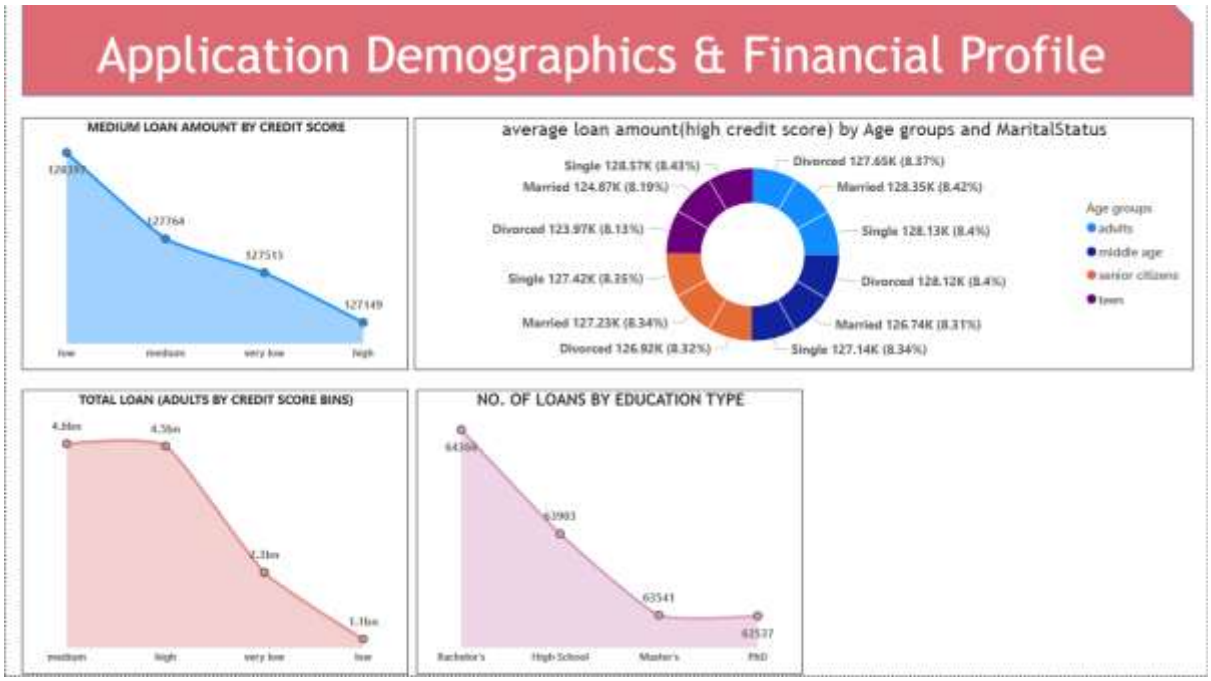
Business Value

This page helps management quickly understand:

- Total exposure to credit risk
- Default trends across the loan portfolio

It acts as the **executive summary dashboard** for decision makers.

Page 2: Applying Demographics & Financial Profile



This page analyzes how **borrower characteristics influence loan repayment behavior**.

Demographic Factors Analyzed

- Age group
- Employment type
- Education background
- Marital status

Financial Profile Indicators

- Income distribution
- Debt obligations
- Loan size relative to income

Insights

The analysis identifies patterns such as:

- Certain demographic groups showing higher default probability
- Borrowers with lower income showing greater financial stress
- Higher debt-to-income ratios correlating with higher default risk

Business Impact

This analysis allows financial institutions to:

- Adjust **credit scoring models**
- Refine **loan approval criteria**
- Improve **risk segmentation**

Page 3: Financial Risk Matrix

This page focuses on **risk classification and financial stability indicators**.

Risk Dimensions

Borrowers are evaluated based on:

- Creditworthiness
- Income stability
- Loan burden
- Repayment capacity

Risk Segmentation

Borrowers are categorized into groups such as:

Low Risk
Moderate Risk
High Risk

Key Insights

The risk matrix highlights:

- Borrowers with high loan amounts but low repayment capacity
- Risk clusters that require closer monitoring
- Segments where default probability is significantly higher

Business Application

Financial institutions can use this matrix to:

- Design **risk-based interest rates**
- Implement **early warning systems**
- Improve **portfolio risk management**

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7. Key Insights from the Analysis

Several patterns emerged from the loan default analysis.

Insight 1

Borrowers with **higher debt relative to income** demonstrate significantly higher default probability.

Insight 2

Certain demographic groups show **greater financial vulnerability**, indicating the importance of demographic segmentation.

Insight 3

Loan amounts exceeding borrower repayment capacity contribute heavily to default cases.

Insight 4

Risk segmentation allows lenders to prioritize **high-risk borrower monitoring**.

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8. Business Recommendations

Based on the insights derived from the analysis, several strategic recommendations can be proposed.

Implement Risk-Based Loan Approval

Loan approval should consider borrower **income stability and debt burden**.

Strengthen Credit Scoring Models

Integrating demographic and financial indicators can significantly improve **default prediction accuracy**.

Introduce Risk-Based Interest Rates

Higher risk borrowers can be charged **higher interest rates** to offset default risk.

Monitor High Risk Borrowers

Institutions should develop **early warning systems** for borrowers showing signs of financial distress.

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9. Business Impact

If implemented in a real financial institution, the insights from this dashboard could help:

- Reduce loan default rates
- Improve credit decision accuracy
- Increase profitability
- Strengthen portfolio risk management

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10. Conclusion

This Power BI project demonstrates how **data analytics can transform financial risk management**. By integrating demographic data with financial indicators, organizations can gain a deeper understanding of borrower behavior and proactively mitigate credit risk.

The dashboard provides an **interactive analytical framework** that supports strategic lending decisions and helps financial institutions build more resilient loan portfolios.